

## SIILINKARI FYR, Finland

WMO: 029430

Lat: 61.5176N

Lon: 23.7539E

Elev: 322

StdP: 14.53

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -10.1      | -4.4 | -14.7                         | 2.5 | -9.6 | -8.6 | 3.5 | -4.0 | 31.9                  | 24.3 | 27.7 | 28.6 | 5.0                   | 0    | 0.709 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.9                   | 78.6            | 65.1 | 75.5 | 63.8 | 72.4 | 62.6 | 68.0                | 74.5 | 66.2 | 73.4 | 63.9 | 70.7 | 9.6                  | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 65.5                            | 95.3 | 71.7 | 63.3 | 88.0 | 70.4 | 60.9 | 80.9 | 67.4 | 32.5          | 74.5 | 31.1 | 73.7 | 29.3 | 70.7 | 73.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 25.8              | 23.0 | 20.8 | -11.0                      | 81.2 | 8.4                | 4.4 | -17.1                                              | 84.3 | -22.0      | 86.9 | -26.7      | 89.3 | -32.8      | 92.5 |
|                   |      |      | -11.4                      | 70.3 | 8.3                | 3.2 | -17.4                                              | 72.5 | -22.3      | 74.4 | -27.0      | 76.2 | -33.0      | 78.5 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 42.0   | 19.9  | 22.7  | 28.2  | 38.3  | 51.0  | 58.2  | 64.2  | 61.5  | 53.5  | 42.3  | 34.9  | 27.9 |
|                                                                               | DBStd       | 16.83  | 11.49 | 11.97 | 8.38  | 6.11  | 7.23  | 6.14  | 6.01  | 4.95  | 5.48  | 5.82  | 6.84  | 9.73 |
|                                                                               | HDD50       | 4222   | 934   | 766   | 675   | 352   | 75    | 6     | 0     | 0     | 30    | 245   | 454   | 685  |
|                                                                               | HDD65       | 8510   | 1399  | 1186  | 1140  | 800   | 437   | 220   | 91    | 132   | 346   | 705   | 904   | 1150 |
|                                                                               | CDD50       | 1296   | 0     | 0     | 0     | 2     | 106   | 251   | 441   | 357   | 134   | 5     | 0     | 0    |
|                                                                               | CDD65       | 109    | 0     | 0     | 0     | 0     | 3     | 15    | 67    | 24    | 0     | 0     | 0     | 0    |
|                                                                               | CDH74       | 435    | 0     | 0     | 0     | 0     | 16    | 67    | 288   | 64    | 0     | 0     | 0     | 0    |
|                                                                               | CDH80       | 50     | 0     | 0     | 0     | 0     | 0     | 4     | 40    | 6     | 0     | 0     | 0     | 0    |
| Wind                                                                          | WSAvg       | 11.1   | 11.0  | 10.9  | 11.1  | 9.6   | 10.3  | 10.7  | 10.0  | 10.5  | 11.2  | 12.7  | 12.4  | 12.5 |
| Precipitation                                                                 | PrecAvg     | 22.3   | 1.4   | 1.0   | 1.1   | 1.3   | 1.4   | 2.0   | 2.8   | 2.9   | 2.4   | 2.2   | 2.1   | 1.7  |
|                                                                               | PrecMax     | 34.0   | 2.9   | 2.6   | 2.4   | 3.5   | 3.5   | 6.7   | 5.2   | 7.2   | 4.4   | 5.1   | 3.8   | 3.6  |
|                                                                               | PrecMin     | 15.8   | 0.5   | 0.2   | 0.0   | 0.0   | 0.3   | 0.6   | 0.8   | 0.9   | 0.2   | 0.5   | 0.4   | 0.3  |
|                                                                               | PrecStd     | 3.8    | 0.6   | 0.6   | 0.6   | 0.8   | 0.7   | 1.3   | 1.3   | 1.5   | 1.0   | 1.1   | 0.9   | 0.8  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 39.0  | 41.7  | 48.4  | 60.5  | 76.6  | 79.9  | 83.9  | 80.4  | 68.8  | 56.6  | 50.0  | 46.2 |
|                                                                               |             | MCWB   | 37.4  | 37.3  | 40.1  | 48.3  | 58.8  | 63.0  | 67.7  | 66.9  | 60.6  | 53.2  | 48.8  | 44.5 |
|                                                                               | 2%          | DB     | 36.8  | 38.4  | 43.1  | 55.5  | 71.1  | 75.5  | 80.3  | 75.0  | 65.5  | 53.7  | 47.5  | 41.6 |
|                                                                               |             | MCWB   | 35.5  | 36.2  | 37.5  | 44.9  | 56.7  | 61.4  | 66.9  | 64.3  | 59.0  | 51.3  | 46.2  | 39.8 |
|                                                                               | 5%          | DB     | 35.0  | 36.5  | 40.3  | 51.2  | 66.9  | 71.5  | 77.0  | 71.8  | 62.7  | 52.3  | 44.9  | 39.3 |
|                                                                               |             | MCWB   | 33.9  | 35.1  | 36.3  | 42.5  | 55.4  | 60.1  | 65.5  | 62.9  | 57.2  | 50.1  | 43.6  | 37.9 |
|                                                                               | 10%         | DB     | 33.5  | 34.8  | 37.8  | 47.5  | 62.9  | 68.2  | 74.1  | 68.9  | 60.7  | 50.3  | 43.0  | 37.7 |
|                                                                               |             | MCWB   | 32.6  | 33.7  | 34.8  | 40.8  | 52.9  | 58.6  | 64.7  | 61.0  | 56.5  | 48.2  | 41.7  | 36.4 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 38.0  | 38.2  | 41.4  | 49.4  | 61.1  | 66.9  | 70.4  | 69.5  | 62.1  | 54.3  | 49.1  | 45.1 |
|                                                                               |             | MCDB   | 38.8  | 39.7  | 46.1  | 58.4  | 69.1  | 73.2  | 78.0  | 76.5  | 66.7  | 56.5  | 49.8  | 46.6 |
|                                                                               | 2%          | WB     | 35.8  | 36.7  | 38.7  | 46.0  | 58.7  | 63.4  | 68.8  | 66.3  | 60.2  | 52.5  | 46.5  | 40.1 |
|                                                                               |             | MCDB   | 36.5  | 38.0  | 42.0  | 52.8  | 69.0  | 71.7  | 76.1  | 72.1  | 63.9  | 53.7  | 47.3  | 41.4 |
|                                                                               | 5%          | WB     | 34.3  | 35.3  | 37.0  | 43.7  | 56.7  | 61.5  | 67.5  | 64.0  | 58.7  | 50.7  | 43.8  | 38.1 |
|                                                                               |             | MCDB   | 35.0  | 36.6  | 39.6  | 49.7  | 64.9  | 69.7  | 74.6  | 70.0  | 61.8  | 51.8  | 44.8  | 39.2 |
|                                                                               | 10%         | WB     | 32.8  | 33.9  | 35.3  | 41.7  | 54.0  | 59.6  | 65.9  | 62.4  | 57.3  | 48.3  | 42.0  | 36.5 |
|                                                                               |             | MCDB   | 33.6  | 34.8  | 37.5  | 46.8  | 61.3  | 66.9  | 73.6  | 67.9  | 60.1  | 50.1  | 43.0  | 37.5 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 9.5   | 9.1   | 12.2  | 12.6  | 13.6  | 11.5  | 10.9  | 9.7   | 8.3   | 6.5   | 5.6   | 6.2  |
|                                                                               |             | MCDBR  | 6.0   | 8.0   | 12.6  | 19.3  | 19.9  | 16.6  | 14.3  | 12.9  | 11.4  | 8.2   | 7.0   | 6.8  |
|                                                                               |             | MCWBR  | 5.9   | 6.9   | 8.9   | 10.7  | 9.0   | 7.3   | 6.2   | 5.8   | 6.2   | 6.5   | 6.7   | 6.6  |
|                                                                               | 5% WB       | MCDBR  | 5.8   | 7.3   | 10.7  | 17.2  | 17.6  | 14.7  | 12.7  | 11.5  | 10.6  | 7.7   | 7.0   | 6.8  |
|                                                                               |             | MCWBR  | 5.8   | 6.5   | 8.1   | 10.3  | 9.1   | 7.1   | 6.5   | 6.2   | 6.4   | 7.0   | 6.8   | 6.7  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.249  | 0.294 | 0.321 | 0.355 | 0.351 | 0.348 | 0.373 | 0.359 | 0.334 | 0.321 | 0.273 | 0.355 |      |
|                                                                               | taud        | 2.084  | 2.167 | 2.253 | 2.377 | 2.448 | 2.481 | 2.430 | 2.460 | 2.522 | 2.413 | 2.183 | 2.596 |      |
|                                                                               | Ebn at Noon | 143    | 211   | 250   | 266   | 276   | 279   | 268   | 262   | 249   | 206   | 134   | 95    |      |
|                                                                               | Edh at Noon | 12     | 22    | 29    | 31    | 31    | 31    | 32    | 29    | 23    | 17    | 11    | 7     |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 69     | 247   | 715   | 1165  | 1585  | 1686  | 1595  | 1247  | 739   | 313   | 81    | 38    |      |
|                                                                               | RadStd      | 13     | 56    | 89    | 133   | 170   | 130   | 161   | 139   | 95    | 60    | 16    | 4     |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (57 neighbors) | +0.88 | N/A     | N/A    | N/A     | N/A   | N/A   | -241        | N/A   | N/A   | N/A   |

Nomenclature: See separate page